

AI TRIAGE SUPPORT SYSTEM DESIGN MVP

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

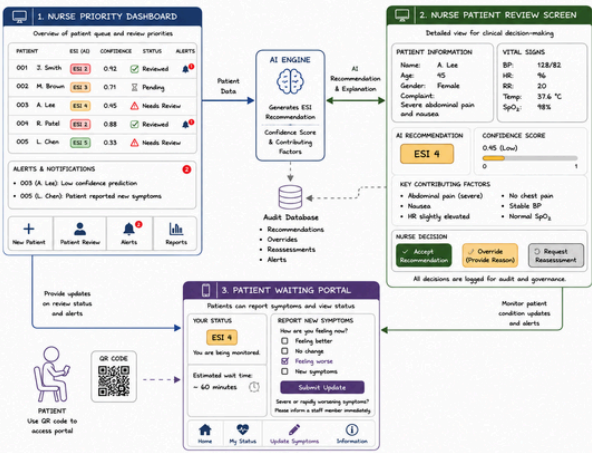
Where and when

- Mercer Emergency Department triage desk and patient waiting area.
- Used during initial triage, reassessment and ongoing patient monitoring.
- Operates continuously (24 hours/day).
- Supports busy periods, overnight shifts and high patient volumes.
- Future expansion to Observation Units and other hospital departments.

Draw a picture

AI TRIAGE SUPPORT SYSTEM - OVERVIEW

HCI Setting A - Mercer ED Triage Desk



AI TRIAGE SUPPORT SYSTEM'S ROLE

- Clinical decision-support assistant.
- Communication facilitator between patients and nurses.
- Prioritisation assistant for patient review.
- Explainable AI system providing advisory ESI recommendations.
- Never replaces professional clinical judgement.

PERSONALITY

- Professional and predictable.
- Calm and non-intrusive.
- Transparent, trustworthy, and consistent across all users.
- Neutral with no emotional behaviour.
- Prioritises patient safety over speed.

CONTEXT-BASED BEHAVIOUR

- Adjust review priority using available clinical information.
- Highlight low-confidence predictions for reassessment.
- Display patient deterioration alerts from the waiting portal.
- Continue supporting manual workflows if data are incomplete.
- Record all recommendations, overrides and reassessments.
- Adapt to future EHR and device integration when available.

CONNECTION TO SYSTEMS

- Mercer Electronic Health Record (future integration).
- Hospital authentication system.
- Clinical audit database.
- Secure QR-code patient portal.
- Future reporting dashboard.
- Secure hospital network.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

- | | | |
|-----------------------------------|---|---|
| ☐ voice | ☐ movement | ☒ screens |
| ☐ sounds | ☒ touch | ☐ lights |
| ☐ gestures | ☐ smell | ☒ other |
| | ☐ facial expressions | |

Keyboard, Mouse,
Manual clinical data entry,
Future EHR import

OUTPUT

- | | | |
|-----------------------------------|---|---|
| ☐ voice | ☐ movement | ☒ screens |
| ☐ sounds | ☐ touch | ☐ lights |
| ☐ gestures | ☐ smell | ☒ other |
| | ☐ facial expressions | |

Color-coded status indicators
Visual alerts, Confidence score
Audit logging

Interaction flow

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

	USER	BEFORE	DURING	AFTER
		• Nurse performs patient assessment. • Clinical information is entered. • Patient receives initial triage.	• Nurse reviews AI recommendation. • Reviews confidence and contributing factors. • Accepts or overrides recommendation. • Patient can submit symptom updates while waiting.	• Final ESI assigned. • Patient monitored. • Alerts reviewed if condition changes. • Audit trail recorded.
ROBOT OPERATOR (optional)	AI SYSTEM	• Waits for patient data. • Validates required inputs.	• Generates ESI recommendation. • Calculates confidence. • Displays contributing clinical factors. • Receives patient updates. • Flags reassessment when needed.	• Records decision history. • Stores override reasons. • Updates audit logs. • Waits for next assessment.